

National University of Computer and Emerging Sciences
Department of Software Engineering

WEB TECHNOLOGY — SE3003
Fall 2026

Semester Project — Phase 1

NexTalk
A Real-Time WebSocket Chat Platform

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Fall 2026

Task 01 — System Overview & Brainstorming

1.1 Project Title

NexTalk — A Real-Time WebSocket Chat Platform

1.2 Problem Statement

In today's fast-paced digital environment, instant communication is no longer a luxury — it is a necessity. Traditional messaging systems rely on polling or page refreshes to deliver messages, introducing significant latency and consuming unnecessary server and bandwidth resources. This approach creates a disjointed experience, particularly in collaborative or team-oriented settings where delays in message delivery disrupt workflow and reduce productivity.

Existing solutions are either too heavyweight for small teams, require paid subscriptions, or lack transparency in how they handle user data. There is a clear need for a lightweight, open, and performant real-time communication platform that uses modern browser technologies to deliver messages instantly, without compromise.

1.3 Proposed Solution Summary

NexTalk is a browser-based real-time chat application built on top of the WebSocket protocol. Unlike HTTP-based polling approaches, WebSockets maintain a persistent, bidirectional connection between the client and server, enabling messages to be pushed instantly as they are sent — with zero unnecessary latency.

The platform supports both one-on-one private messaging and group chat rooms. Users can register, log in, and instantly begin communicating through a clean and intuitive interface. An administrative panel allows authorized personnel to manage users, monitor activity, and moderate content.

NexTalk is designed with scalability and simplicity in mind. The front end is implemented using HTML, CSS, and JavaScript, while the back end leverages Node.js with the ws library to manage WebSocket connections. All communication is handled through defined message events, ensuring a structured and maintainable codebase throughout the development lifecycle.

Task 02 — Core Features, Modules & Actors Identification

2.1 Actors

Primary Actors:

- Registered User — The main consumer of the platform. Creates an account, logs in, sends/receives messages in real time, manages profile, and participates in group chats.
- Administrator — Manages and monitors the platform. Can view all users, disable accounts, delete inappropriate messages, and view system analytics.

Secondary Actors:

- Guest / Unregistered Visitor — Can view the landing/marketing page and initiate registration but cannot access chat features.
- WebSocket Server — An internal system actor that manages all real-time connection events: connection, disconnection, message broadcasting, and presence tracking.
- Database System (MongoDB / MySQL) — Stores user credentials, chat histories, room metadata, and user profile information.

2.2 Key Features

#	Feature	Description
1	Real-Time Messaging	Instant message delivery using WebSocket persistent connections with zero polling overhead.
2	User Authentication	Secure registration and login with hashed passwords and session management.
3	Group Chat Rooms	Users can create, join, and leave named chat rooms. Rooms support unlimited members.
4	Private Messaging	One-on-one direct message threads between any two registered users.
5	Online Presence Indicators	Live display of which users are currently online within a room or across the platform.
6	Message History	Persistent storage of chat logs so users can scroll back through previous conversations.
7	User Profile Management	Users can update display name, avatar, and status/bio from a profile settings page.
8	Admin Dashboard	Centralized panel for user management, content moderation, and platform-level analytics.

9	Typing Indicators	Real-time 'User is typing...' feedback broadcast over WebSocket to all room participants.
10	Notifications	In-app notification badge and alert when a new message arrives in an inactive tab or room.

2.3 System Modules

Module 1 — Authentication Module

Purpose: Handles all aspects of identity management including new user registration, login, session creation, logout, and password recovery. Ensures that only verified users access the platform.

Actors: Registered User, Guest/Unregistered Visitor

Module 2 — Real-Time Messaging Module

Purpose: The core of the application. Manages WebSocket connections, handles message events, broadcasts messages to appropriate rooms or users, and tracks typing activity and online presence.

Actors: Registered User, WebSocket Server

Module 3 — Chat Room Management Module

Purpose: Allows users to create new group chat rooms, browse available public rooms, join or leave rooms, and view room membership lists. Stores room metadata in the database.

Actors: Registered User, Database System

Module 4 — User Dashboard Module

Purpose: The central hub users land on after login. Displays recent conversations, active rooms, friend/contact list, notifications, and quick-access controls to start a new chat.

Actors: Registered User

Module 5 — Profile Management Module

Purpose: Enables users to update personal information including display name, profile photo, account password, and status message. Displays activity statistics such as messages sent.

Actors: Registered User

Module 6 — Admin Panel Module

Purpose: Restricted to administrators. Provides a comprehensive dashboard showing system metrics, a full user list with enable/disable controls, message moderation tools, and room oversight.

Actors: Administrator, Database System

Task 03 — Screen / Interface Planning

Screen 1 — Landing / Home Page

URL Route	/ (index)
Accessible By	All visitors (Guest + Registered)
Information Displayed	App branding, tagline, key feature highlights, screenshots/mockup preview, and call-to-action buttons.
User Actions	Click 'Get Started' (redirects to Register), click 'Login' (redirects to Login page), scroll through feature overview.

Screen 2 — Registration Page

URL Route	/register
Accessible By	Guest / Unregistered Visitor
Information Displayed	Registration form with fields: Full Name, Username, Email, Password, Confirm Password. Link to Login page.
User Actions	Fill in form fields, submit to create account, toggle password visibility, navigate to Login if already registered.

Screen 3 — Login Page

URL Route	/login
Accessible By	Guest / Unregistered Visitor
Information Displayed	Login form with Username/Email and Password fields. 'Remember Me' checkbox, link to Register and Forgot Password.
User Actions	Enter credentials, submit form, toggle password visibility, navigate to Register or Forgot Password.

Screen 4 — User Dashboard

URL Route	/dashboard
Accessible By	Registered User (authenticated)

Information Displayed	Left sidebar: chat room list, direct message contacts, online indicators. Center panel: active chat window with message history. Right sidebar: room member list, user profiles.
User Actions	Send and receive real-time messages, switch between rooms/DMs, view typing indicators, access profile settings, start a new chat, view notifications.

Screen 5 — Chat Room Page

URL Route	/room/:roomId
Accessible By	Registered User (room member)
Information Displayed	Room name and description at top, scrollable message history with sender avatars/names and timestamps, 'X is typing...' indicator, online member count.
User Actions	Type and send messages, view/scroll message history, see who is typing, leave the room, view member list.

Screen 6 — Rooms Browser Page

URL Route	/rooms
Accessible By	Registered User
Information Displayed	Grid/list of all available public chat rooms with room name, description, member count, and creation date.
User Actions	Search/filter rooms by keyword, join a room, create a new room (opens creation modal with name/description inputs).

Screen 7 — User Profile Page

URL Route	/profile
Accessible By	Registered User (own profile) or viewing others
Information Displayed	Profile photo, display name, username, status/bio, date joined, message count statistics. Edit controls visible only on own profile.
User Actions	Update display name, bio, upload new avatar, change password, view own stats, send DM to another user (when viewing others' profiles).

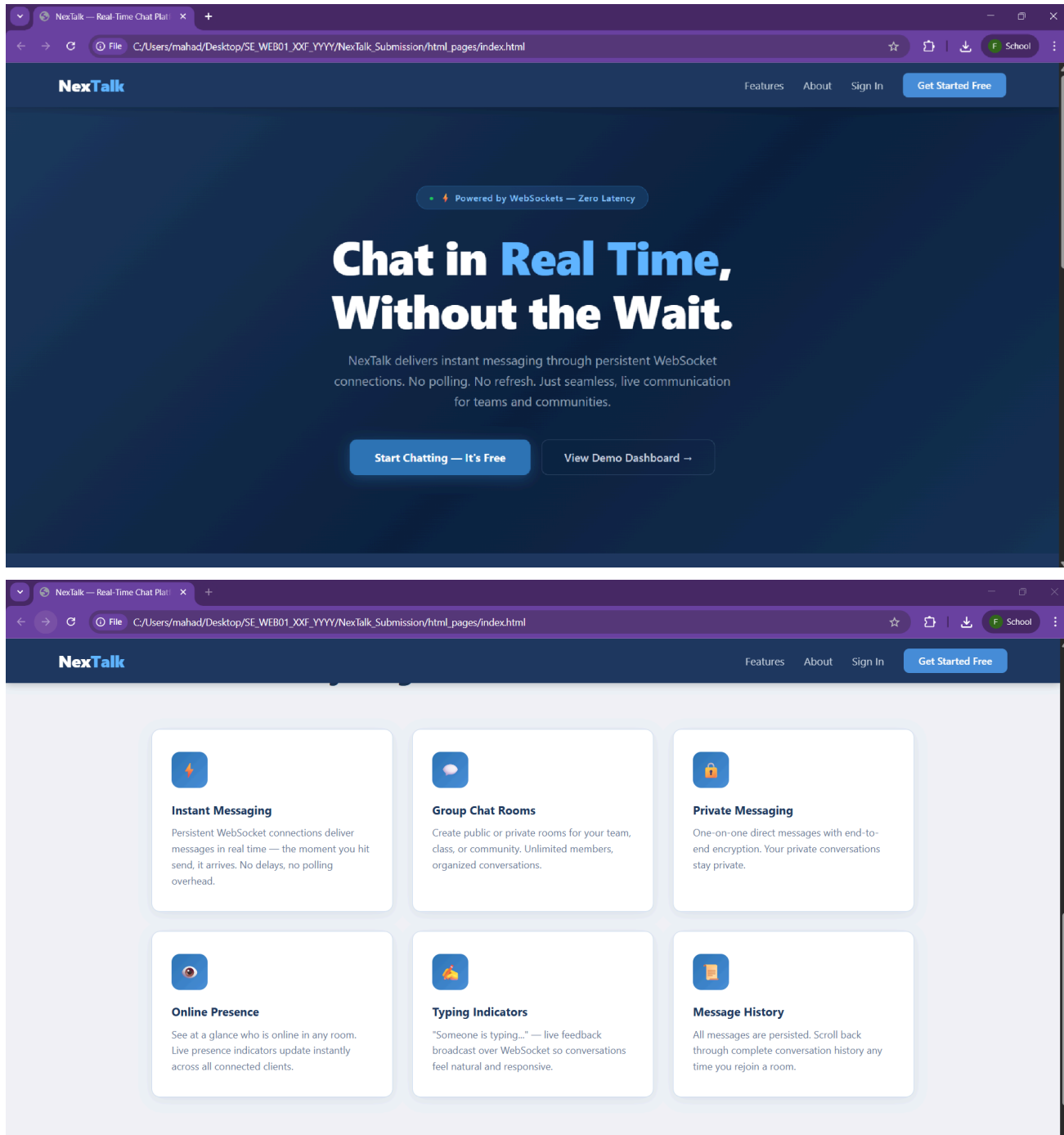
Screen 8 — Admin Panel

URL Route	/admin
Accessible By	Administrator only
Information Displayed	Summary cards (total users, active rooms, messages today), full user table with account status, room management list, recent flagged messages log.
User Actions	Enable/disable user accounts, delete rooms, remove flagged messages, filter user list, export user data.

Task 04 — Initial Page Development (HTML + CSS)

Three web pages have been implemented as part of this phase using HTML5 and CSS3. The pages developed are:

- Page 1 — Landing / Home Page (index.html + style.css)



- Page 2 — Login & Registration Page (auth.html + style.css)

NexTalk
Real-Time Chat — Powered by WebSockets

[Sign In](#) [Create Account](#)

Email or Username
you@example.com

Password

☐ Remember me for 30 days [Forgot password?](#)

[Sign In →](#)

Don't have an account? [Create one free](#)

NexTalk
Real-Time Chat — Powered by WebSockets

[Sign In](#) [Create Account](#)

First Name Last Name
Ali Hassan

Username
alihassan123

Email Address
ali@example.com

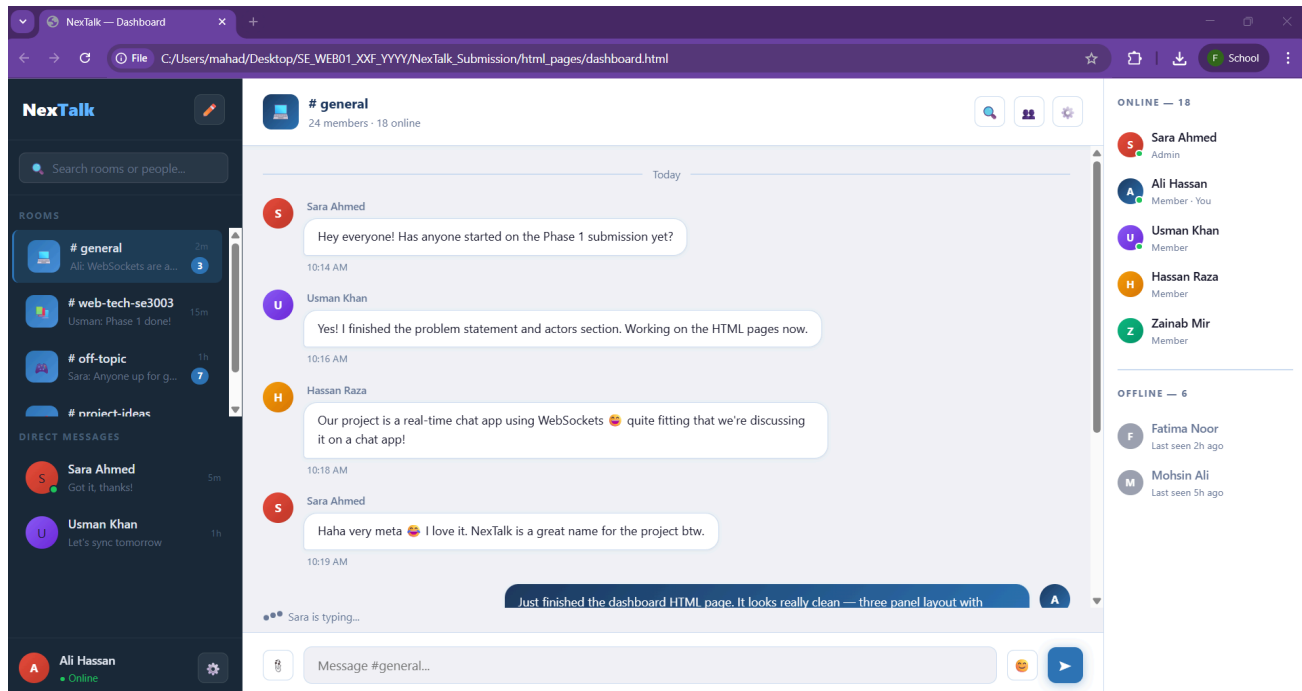
Password
At least 8 characters

Confirm Password
Re-enter password

[Create My Account →](#)

Already registered? [Sign in here](#)

- Page 3 — Chat Dashboard (dashboard.html + style.css)



All pages share a common stylesheet (style.css) and implement the following design standards:

- Responsive layout using CSS Flexbox
- Consistent color palette: Deep Navy (#1E3A5F), Sky Blue (#2E75B6), Light Backgrounds
- Clean typography using the system-safe 'Segoe UI' / sans-serif font stack
- User-friendly spacing, card-based UI components, and accessible form elements
- Hover effects and smooth CSS transitions for interactive elements

The HTML and CSS source files are included alongside this document in the project submission folder. Screenshots of each page are embedded below for reference.

Implementation Notes:

- The landing page features a hero section with a call-to-action, a three-column features section, and a footer.
- The authentication page contains both a Login form and a Registration form, toggled via a tab-based interface using CSS and minimal JavaScript.
- The dashboard page demonstrates the three-panel chat interface: left sidebar (rooms/contacts), center chat window, and right member panel.